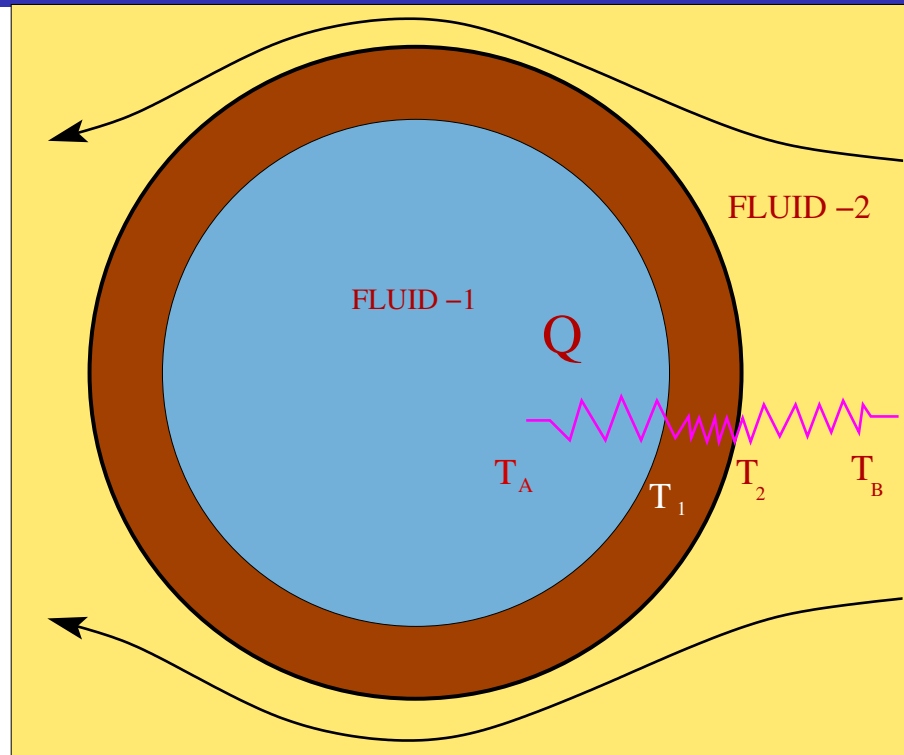


Overall Heat Transfer Coefficient



Heat transfer through tube

- Heat transfer from Fluid A to Tube material
$$Q = h_A A_i (T_A - T_1),$$
- h_A - Tube inside convection heat transfer coefficient, Tube inside surface area $A_i = \pi D_i L$

- Heat transfer through the tube material $Q = \frac{(T_1 - T_2)2\pi kL}{\ln\left(\frac{r_o}{r_i}\right)}$
- Heat transfer from Tube to Fluid B, $Q = h_B A_o (T_2 - T_B)$,
- h_B - Tube outside convection heat transfer coefficient, Tube outside surface area $A_o = \pi D_o L$
- From the above three eqns $Q = \frac{T_A - T_B}{\left(\frac{1}{h_i A_i} + \frac{\ln\left[\frac{r_o}{r_i}\right]}{2\pi kL} + \frac{1}{h_o A_o}\right)}$
- Heat transfer $Q = \frac{A_i (T_A - T_B)}{\left(\frac{1}{h_i} + \frac{A_i \ln\left[\frac{r_o}{r_i}\right]}{2\pi kL} + \frac{A_i}{h_o A_o}\right)}$
- We define U_i The overall heat transfer coefficient based on inside surface area as $U_i = \frac{1}{\left(\frac{1}{h_i} + \frac{A_i \ln\left[\frac{r_o}{r_i}\right]}{2\pi kL} + \frac{A_i}{h_o A_o}\right)}$
- The heat transfer $Q = U_i A_i (T_A - T_B)$

- Heat transfer $Q = \frac{A_o(T_A - T_B)}{\left(\frac{A_o}{A_i h_i} + \frac{A_o \ln \left[\frac{r_o}{r_i} \right]}{2\pi k L} + \frac{1}{h_o} \right)}$
- We define U_o The overall heat transfer coefficient based on outside surface area as $U_o = \frac{1}{\left(\frac{A_o}{A_i h_i} + \frac{A_o \ln \left[\frac{r_o}{r_i} \right]}{2\pi k L} + \frac{1}{h_o} \right)}$
- The heat transfer $Q = U_o A_o (T_A - T_B)$

- Heat transfer $Q = \frac{A_o(T_A - T_B)}{\left(\frac{A_o}{A_i h_i} + \frac{A_o \ln \left[\frac{r_o}{r_i} \right]}{2\pi k L} + \frac{1}{h_o} \right)}$
- We define U_o The overall heat transfer coefficient based on outside surface area as $U_o = \frac{1}{\left(\frac{A_o}{A_i h_i} + \frac{A_o \ln \left[\frac{r_o}{r_i} \right]}{2\pi k L} + \frac{1}{h_o} \right)}$
- The heat transfer $Q = U_o A_o (T_A - T_B)$
- U_o and U_i different ?, what happens to A_i and A_o ?

Fouling in Heat exchangers



Fouling

- Fouling - due to deposition of dirt, mud, contaminants, corrosion, erosion, etc.
- Fouling on both sides -inside the tube and outside the tubes.
- Deteriorate heat exchanger performance.

Fouling considerations - Design

- Fouling factor (K/W)
- Fouling factor to be considered during design ?
- High or Low value for Fouling factor ?
- Area requirement with fouling considerations
- Overall heat transfer coefficient with Fouling factor
- How to get fouling factor values ? [► Foul Fact](#)
- Velocity considerations

Fouling in Heat exchangers

Typical Fouling Factors [$\text{m}^2\text{K/W}$]

Cooling Water

Conditions

cooling water < 50 C
cooled fluid < 120 C

cooling water > 50 C
cooled fluid > 120 C

Cooling Water velocity

$v < 1 \text{ m/s}$

$v > 1 \text{ m/s}$

$v < 1 \text{ m/s}$

$v > 1 \text{ m/s}$

Type of Water

Sea	0.00009	0.00009	0.00018	0.00018
Brackish	0.00035	0.00018	0.00053	0.00035
Cooling tower with inhibitor	0.00018	0.00018	0.00035	0.00035
Cooling tower without inhibitor	0.00053	0.00053	0.00088	0.0007
City grid	0.00018	0.00018	0.00035	0.00035
River minimum	0.00018	0.00018	0.00035	0.00035
River average	0.00053	0.00035	0.0007	0.00035
Engine jacket	0.00018	0.00018	0.00018	0.00018

Fouling

- Type of fluid, Fluid velocity, Temperature
- Turbulators, Service conditions

Overall Heat transfer coefficient U

- Overall heat transfer coefficient

$$U_i = \frac{1}{\left(F_i + \frac{1}{h_i} + \frac{A_i \ln \left[\frac{r_o}{r_i} \right]}{2\pi kL} + \frac{A_i}{h_o A_o} + \frac{F_o A_i}{A_o} \right)}$$

- Overall heat transfer coefficient

$$U_o = \frac{1}{\left(\frac{A_o F_i}{A_i} + \frac{A_o}{A_i h_i} + \frac{A_o \ln \left[\frac{r_o}{r_i} \right]}{2\pi kL} + \frac{1}{h_o} + F_o \right)}$$